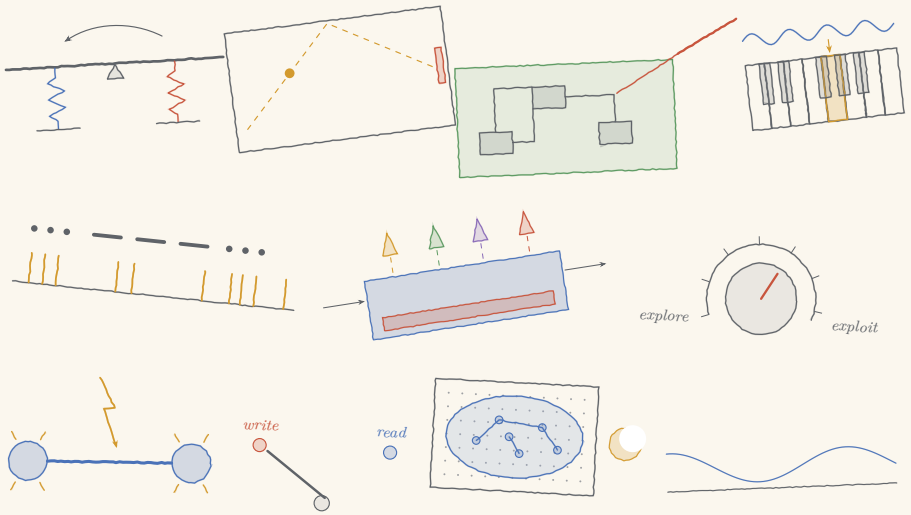




The 20-Watt Computer

How the Brain Computes, Short and Simple

Konstantin Kladko



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Translated from the Russian edition.

The full version, sources and models: <https://github.com/kladkogex/20-watt-computer>.

Contents

1	A Bag of Parts	1
2	A Machine Without a Conductor	6
3	Brakes	11
4	Morse Code	15
5	Talking Through the Air	19
6	How the Machine Learns	23
7	What Dopamine Really Says	27
8	The Adventure Knob	31
9	Write and Read	35
10	The Whole Machine	38
11	How the World Gets In	41

12 How to Recognize a Cat	45
13 The Only Fast Output	49
14 The Alarm Signal	53
15 How the Machine Learns to Move	56
16 The Second Output	59
17 How to Fix a Machine Without a Schematic	62
18 A Neuron Is Not One Device	65
19 How Many Inputs Are Needed	67
20 What the Machine Does at Night	70
21 A Computer Made of Living Cells	74
22 Neurons Play Tennis	78

Before We Begin

This book is about a machine that weighs a kilogram and a half, runs on twenty watts, like a dim light bulb, and still recognizes faces, learns languages, and catches a flying ball. Your brain. We will look at it the way an engineer looks at an unfamiliar circuit board: what the parts are, how they are connected, what signals run along the wires, and what the circuit computes.

You will need neither formulas nor any knowledge of biology. You will need curiosity and a willingness to picture a wire, a light bulb, a switch. We will pick up everything else along the way.

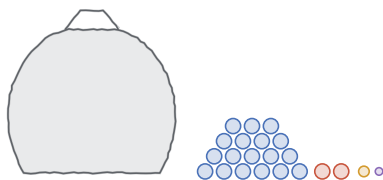
The book has a second, full version, with equations, models, exercises, and references to the experiments. The chapters of the two versions come in the same order and carry the same numbers. If a chapter hooks you, open its full version: there you will find where each claim comes from and how to check it yourself. On the book's website every chapter switches between "Short" and "Full" with a single button.

One warning. We do not know everything about the brain, far from it, and I will say honestly where the measurements end and the guesswork begins. This is not a flaw of the book but the most interesting thing in it: many chapters end with

questions nobody can answer yet.

Chapter 1

A Bag of Parts



A bag of parts: almost all blue, a few red, and a handful of others.

Imagine you are handed a bag holding eighty-six billion parts that all look alike, and asked to work out how a computer is built from them. Where would you start? An engineer would not study the shape of the parts. An engineer would ask: how many inputs does a part have, how many outputs, and what does it put out?

The neuron, as such a part is called, is surprisingly simple. It has thousands of inputs: wires from other neurons run to it. It has one output. But that output branches like a tree, and the same copy of the signal goes out to thousands of receivers. Picture a person who listens to a thousand people

at once and answers with a single sentence, but in such a way that another thousand people hear it.

What is this signal? A short click. Every now and then the neuron fires a spike, a voltage blip about a thousandth of a second long. All the clicks of one neuron are the same, like the beats of one drum. So everything a neuron can say is written not in how loud it clicks but in *when* it clicks: more often, less often, at once, or after a delay.

Inside, the neuron performs the simplest computation imaginable: it adds up the incoming signals and, if the sum goes over a threshold, clicks itself. Some inputs push the sum up, others push it down.

How to Sort the Bag

When the tip of the output wire reaches a neighbor, it does not touch it. A narrow gap remains between them, and chemistry carries the signal across: the wire releases a pinch of a special substance, a transmitter, and the neighbor catches it. And here is a stroke of luck: almost every neuron releases the same substance at all the tips of its wire. So we can sort neurons by *what* they release.

We will name a neuron type by the first four letters of its substance. A neuron that releases glutamate is **GLUT**. One that releases gamma-aminobutyric acid is **GABA**. Then come **DOPA** (dopamine), **SERO** (serotonin), **NORA** (nora-drenaline), **ACET** (acetylcholine), and a few more.

Once the bag is sorted, the picture is very lopsided. Out of every thousand neurons about nine hundred sixty are **GLUT**. Their signal nudges the receiver toward a click: this is a “plus.” About thirty-six more are **GABA**; their signal pushes the receiver away from the threshold: this is a “minus.”

In the cortex there are more of them, from a fifth to a quarter. And all the other types together come to a few neurons per hundred thousand. A drop in the ocean.

Telephone and Radio

But this drop is built quite differently. The wire of a **GLUT** or **GABA** neuron leads to particular neighbors, like a telephone line: only those who are called get the signal. The tiny groups of **DOPA**, **SERO**, **NORA**, and **ACET** neurons, by contrast, work like radio stations. Their wires spread over huge regions of the brain, the substance is often released not into a gap but into the space around it, and millions of cells hear the same message at once.

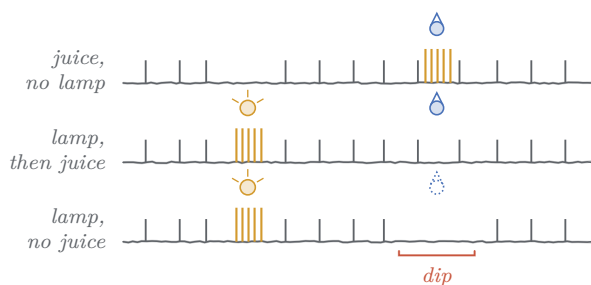
The telephone lines carry data: here is a line in the picture, here is a sound of a certain pitch. The radio carries global settings: how loud everything is, whether it is worth learning, whether it is time to sleep. Any computer is the same: there are billions of memory cells and a handful of control registers. Without them the machine does not work anyway.

The Receiver Chooses the Meaning

There is one subtlety. The substance itself does not know whether it is a “plus” or a “minus.” The molecule is a key. What happens when the key is inserted is decided by the lock, a special protein on the receiver’s side. The same dopamine perks up some receivers and damps others, depending on which lock they carry. It is like one radio broadcast that different listeners take in different ways.

The “Better Than Expected” Signal

The most famous radio channel is dopamine. What does it broadcast? Here is an experiment. An animal is unexpectedly given a drop of juice, and the dopamine neurons answer with a flurry of clicks. Then a lamp is lit before every juice. The animal learns that the lamp promises juice, and now the flurry comes at the lamp, and not at the juice itself. And if no juice follows the lamp, the dopamine neurons fall *silent* at the moment it was expected.



A flurry at unexpected juice; at the lamp that promises it; and a dip at the moment the promised juice fails to come.

One rule explains all three cases: dopamine reports not “good” but “better than I expected.” Unexpected juice is a pleasant surprise. Promised juice is no surprise at all. Promised but not given is an unpleasant one. Programs that learn from their mistakes use exactly this quantity. We will come back to it.

What to Take Away

The brain is a huge network of parts that all look alike. Almost all of them are the “pluses” and “minuses” of the

telephone network of data. And very few are radio stations that tune the whole network at once. In the next chapter we will see what can be built from “pluses” alone.

Formulas, models, exercises, and sources are in the full version, Chapter 1.

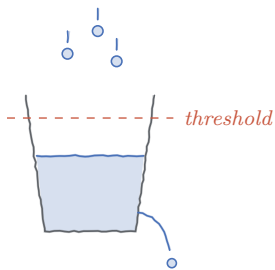
Chapter 2

A Machine Without a Conductor

Your computer has a conductor: the clock generator. Billions of times a second it gives the beat, and all the elements of the circuit switch at once, in step. The brain has no conductor at all. Each neuron works on its own: signals arrive at it whenever they happen to, and leave it whenever they happen to. Let us see what such a machine can compute if we take only **GLUT** neurons, the “pluses” alone.

The Leaky Bucket

It helps to picture a neuron as a bucket with a hole. Each incoming spike is a scoop of water. The water slowly drains out through the hole. If scoops arrive often, the water has time to rise to the brim, and the bucket tips over: the neuron clicks, the bucket is empty, and everything starts again. If scoops arrive rarely, the water has time to drain, and nothing happens.



A neuron is a leaky bucket: drops of input fill it, the water drains away; when it reaches the line, a click.

The hole is the main difference between a neuron and a logic gate. A neuron does not remember an incoming spike forever, only for some milliseconds. Everything that arrives within this window adds up; whatever came earlier is forgotten.

“Or” and “And”

If one scoop is enough to tip the bucket, the neuron fires on any input. This is “or.” If two scoops are needed, things get interesting: without a conductor, “did both arrive?” makes no sense until we say *within what time*. The neuron fires only if the two spikes arrive almost together, before the first scoop drains away. This is “and in time,” a coincidence detector. For some neurons the window is a few milliseconds; for hearing neurons, a fraction of a millisecond.

What “Pluses” Alone Cannot Do

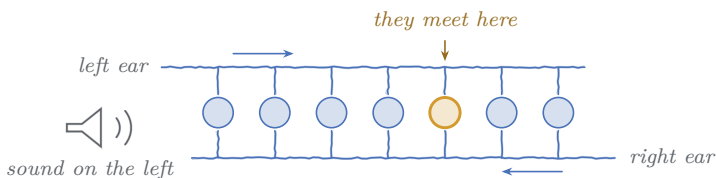
Try to build “not”: a neuron that works when its input is silent. It will not work: silence pours no water into the bucket. And this holds not only for one neuron but for any

network of “pluses” alone: if you strengthen the inputs, no neuron in such a network will become less active. Hence the nastiest property: *activity cannot be put out by activity*. A fire that has flared up can be stopped in only one way: stop throwing on wood.

Nature found a way around. In many sense organs the same stimulus is carried by two groups of neurons. In the eye some cells report “it got brighter,” others “it got darker.” If instead of one wire you have a pair of wires, “yes” and “no,” then “not” comes for free: just swap the wires. With such pairs you can build any logic circuit from “pluses” alone, and even tell that the computation is finished: in every pair exactly one wire has lit up. Engineers build asynchronous circuits this way, circuits that work correctly with any delays.

Time from Delays

Without a conductor there is no common clock, but time can still be computed. Sound from a source off to one side reaches the near ear earlier than the far one, by a fraction of a millisecond. Let us run a wire from the left ear left to right along a row of coincidence detectors, and one from the right ear right to left. The two signals run toward each other and meet at the point both reach at the same moment. The detector at exactly that spot fires. The time difference has turned into the *number* of the neuron that fired, that is, into the direction of the sound. The precision is about ten microseconds, even though each single neuron is far less precise: what saves the day is that there are many detectors.



Sound from the left reaches the left ear first, its signal gets farther along, and the meeting lands to the right of the middle. The number of the neuron that fires is the direction of the sound.

Memory Is a Campfire

Take a group of neurons that strongly excite one another. If you light it with a brief signal, it will keep itself going, like a campfire that feeds its own flames. The signal is over, but the group keeps burning: it remembers that the signal was there. This is a memory cell. And if the signal was too brief, the fire has no time to catch and dies out.

But here is the trouble: how do you erase such a memory? With activity, there is no way. All that remains is to wait until the neurons tire and the fire burns out.

A Timer Started by an Event

Let us chain the groups: the first lights the second, the second the third. Push the first, and a wave runs down the chain like a row of falling dominoes. The number of the fallen domino tells how much time has passed since the push. This is not a clock that always ticks but a timer that is switched on by an event and runs once. In songbirds, during a song, the neurons of one region fire strictly in turn, much like such a chain.

So a lot can be built from “pluses” alone. Three things are missing: picking one out of many, erasing memory on command, and keeping the network from catching fire. All three come from the “minuses,” the subject of the next chapter.

Formulas, models, exercises, and sources are in the full version, Chapter 2.

Chapter 3

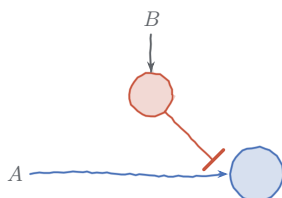
Brakes

Let us add the second most numerous type of neuron to the machine: **GABA**. Its signal does not nudge the neighbor toward a click; it pushes it away. These are the brakes. There are not many of them, from a fifth to a quarter of all neurons in the cortex, but without them the machine does not run.

There is one limitation: a “plus” neuron cannot brake a neighbor by itself. If you need a negative connection, you have to build it through a go-between: the “plus” wakes up a braking neuron, and that one brakes the target. Flipping the sign costs one neuron and a small delay.

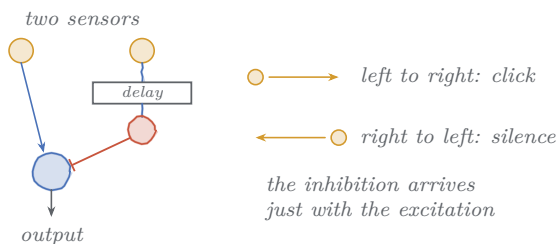
“But Not If”

The most useful operation with brakes is a veto: “fire on A, but not if B has come.” The neuron gets a push from A and a brake from a go-between that B wakes up. Vetoes and “or” are enough to build any logic.



“A, but not if B”: the red neuron blocks the blue one’s path.

A veto in time gives the motion detector that sits in your eye. Two neighboring sensors: one excites the output neuron, the other inhibits it with a delay. If a spot of light runs in one direction at the right speed, the inhibition arrives exactly when the excitation does and cancels it. If it runs the other way, the inhibition is late, and the neuron manages to click. The result is a neuron that sees motion in one direction only.

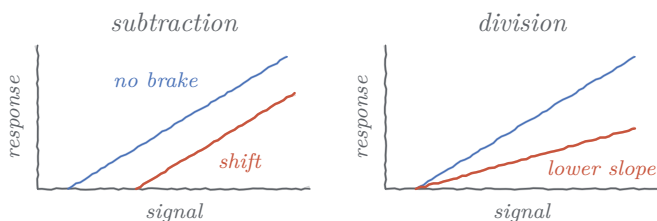


A motion detector: the inhibition arrives with a delay and cancels the response only if the spot runs from right to left.

Subtraction or Division

Inhibition can do more than subtract. An open braking channel does not pull the voltage down; it pulls it *toward rest*,

as if a second hole had been punched in the bucket. Because of this, any inflow of water raises the level less. Inhibition does not take a fixed portion away from the signal; it *divides* it: this is a volume control. If the strength of inhibition grows with the overall activity around, each neuron responds not to how strong its own signal is but to how strong it is *compared with its neighbors*. That is why the same network works both in dim light and in bright sunshine. True, such a “divider” acts cleanly on the spike rate only together with steady background noise, which is always present in a living brain.



Blue is the neuron's response without inhibition, red is with it. Subtraction shifts the threshold; division turns down the volume, like a control knob.

There is a second consequence: the extra hole makes the bucket forget faster. The window within which two signals must land to add up gets narrower. Inhibition makes coincidence detectors stricter.

Choice

Now for the main thing that was missing: choosing one out of many. Two groups of neurons inhibit each other. While the inhibition is weak, both work, just more quietly. But if it is strong enough, any small difference grows, as on a

seesaw, until one group falls completely silent. Winner takes all. And the win holds: for the network to change its mind, a new argument must be noticeably stronger than the old one. The machine does not jump at every rustle.

Brakes also erase memory: on a “reset” signal a braking neuron puts out the campfire from the last chapter. Two such cells that inhibit each other are the counterpart of a latch, the basis of all computer memory.

Life on the Edge

A network of “pluses” alone can run away into an avalanche: each spike gives rise to more than one next spike. The brakes hold it at its operating point, the way a thermostat holds the temperature of an oven. There are two conditions: the inhibition must be strong enough and fast enough. If it is strong but slow, the regulator starts to overshoot, like someone twisting the shower tap without waiting for the hot water to arrive. The temperature swings back and forth. In the brain such a loop of “pluses” and “minuses” produces a fast rhythm, several dozen oscillations per second.

The cortex, it seems, lives right on the edge: its own feedback is strong enough to run away into an avalanche without the brakes. Only the fast brakes hold it back, and that is why it responds so sensitively to weak signals.

The chapter’s conclusion is unexpected: brakes add almost no new computations to the machine. They add *control*: choice, reset, volume control, stability. Excitation carries the data; inhibition decides which data get through, when, and how loudly.

Formulas, models, exercises, and sources are in the full version, Chapter 3.

Chapter 4

Morse Code

Morse code has two signs, the dot and the dash, and the pauses between them. A neuron's alphabet is even poorer: it has one sign, a short click. So the whole message is written in the pauses. What language, then, do neurons speak?

How Much Clicks Can Say

Let us start like a communications engineer: how much information fits into such a channel at all? If the receiver can tell the arrival time of a click to within a millisecond, one click can carry about eight bits. If the receiver only counts how many clicks arrived in a second, it will extract dozens of times less from the same stream. Almost all the capacity of the channel lies in *timing*.



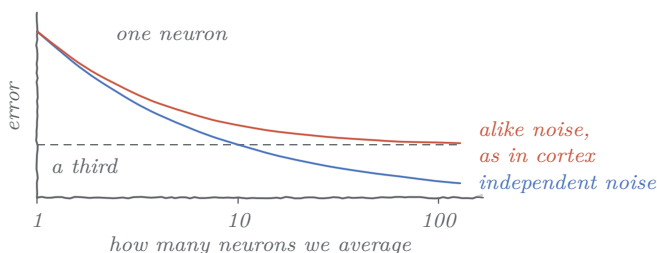
*Morse code above, a neuron below: one sign, the click;
the whole message is written in the pauses.*

But the channel is noisy. The neuron itself, oddly enough, is precise: feed it the same varying current many times, and it will click at the same moments to within a millisecond. The noise is in the connections: more than half of the clicks that reach the end of the wire are simply lost, because the substance is not released every time. In the living cortex the stream of clicks looks almost random. Is it noise, or a message we cannot read yet? A well-compressed message looks random to an outsider, so the question is open.

Slow but Sure

The simplest code is to count clicks: the more often, the larger the number. Losses and jitter do not hurt this code. But it is slow. To know a rate to within ten percent you need a hundred clicks, and that takes seconds. Yet you recognize an animal in a picture flashed for an instant in about a hundred and fifty milliseconds, and on the way the signal passes about ten stages of processing. At each stage each neuron has time to click once at most.

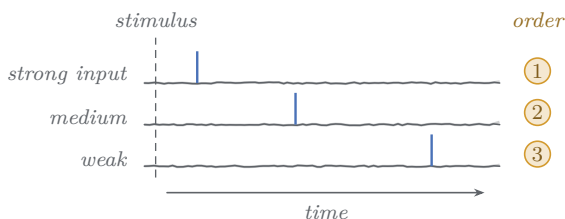
The way out is to take many neurons at once: not one listener counting clicks for a minute, but a hundred listeners for an instant. But there is a catch here too: the noise of neighboring neurons is a little alike, like rumors in a crowd. Averaging over a few dozen neighbors helps, and beyond that it almost stops helping.



The error of the rate averaged over a group of neurons. If the noise were independent, it would keep falling. The alike noise of neighbors stops it at about a third, and this limit is reached already at a few dozen neurons.

Fast: Who Comes First

The second way out is to write the number in time. A strong signal makes a neuron click early, a weak one late. A single click already carries a number. But what do we count time from? We can compare neurons with one another: who clicked first, who second. The order in which ten neurons click carries more than twenty bits. And a common volley serves as the start mark, like the long pause between words in Morse code.



A strong input gives an early click, a weak one a late click. The order of the clicks tells which input is stronger, even when the starting moment is unknown.

The Listener Chooses the Code

One and the same sequence of clicks carries the rate, the timing, and the order. Which of these gets read is up to the receiver. A neuron with a “wide hole” in its bucket slowly accumulates water and hears only the rate. A neuron that forgets quickly hears only coincidences. And the brakes, as we saw, can switch a neuron from one mode to the other.

Even the connections filter the signal. Some tire quickly and so report not the rate but its *change*. Others, on the contrary, pass bursts of several clicks in a row better: the neuron gains a “dash.” Meanwhile the braking neurons put in the punctuation: they cut the stream into windows and turn it down when it gets too loud.

A Thrifty Language

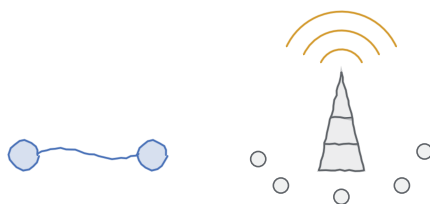
And finally: this language is expensive. Each click costs energy, and the whole brain consumes about twenty watts. Divide those watts among all the neurons, and you get on average about one click per second per neuron. Most neurons are silent most of the time. The brain’s code has to be stingy.

In Morse code the most common letter of English, E, is a single dot: frequent things are coded short. Neurons are similar: what is expected costs few clicks. A constant stimulus gradually stops evoking a response, sensors report changes, and dopamine reports only surprises. The brain spends its clicks on news.

Formulas, models, exercises, and sources are in the full version, Chapter 4.

Chapter 5

Talking Through the Air



On the left, a telephone: a wire from one to one. On the right, a radio: one message to everyone around at once.

So far the neurons have talked to one another by telephone: wire to wire, addressed. Now let us switch on the first radio station: the **SERO** neurons, which release serotonin. There are only a few hundred thousand of them in the whole brain, but their wires spread over huge regions of it.

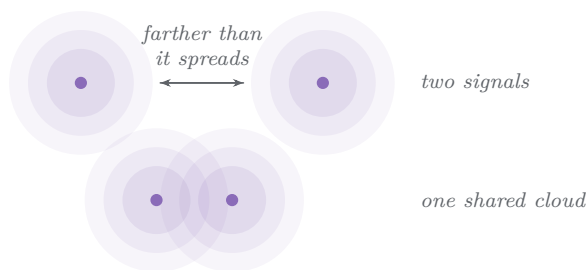
Four Differences

A **SERO** neuron is unlike a **GLUT** neuron. First, the sign of its signal is chosen by the receiver: serotonin has “locks” of both signs. Second, it runs by itself: when we are awake, it clicks steadily a few times a second, as long as it gets a constant background supply. It is a metronome. Third, it is slow: its signal acts not in milliseconds but over hundreds of milliseconds. Fourth, it often releases its substance not into a narrow gap but into the space around, where it spreads like a drop of ink in water. The receiver senses the concentration and does not know which neuron a molecule came from.

Logic That Isn’t There

A metronome with a braking “lock” is a ready-made “not”: while there is no serotonin, the neuron works; once it appears, the neuron falls silent. One neuron instead of a whole circuit. It seems the serotonin network is logically stronger than the **GLUT** network.

But there is a snag. A substance released into space gets mixed. Two signals stay distinct only if they are released at places farther apart than the substance has time to spread, and that is tens of micrometers. Each serotonin neuron, however, has a huge number of release sites scattered over large regions, and the “wires” of different neurons overlap. You cannot send separate bits through such a network. It can carry only a few global quantities, like a radio on which everyone listens to the same station.



Serotonin is released into space and spreads as a cloud. Two sources give two different signals only if they are farther apart than the substance has time to spread. If they are closer, the clouds merge into one shared level.

What It Really Does

For one global quantity this system has everything it needs. The serotonin concentration smooths the separate clicks into a smooth level, the way the simplest converter turns a stream of pulses into a voltage. This level holds for seconds and fades by itself: not a bit that has to be erased, but a slowly vanishing trace.

Serotonin neurons also carry “locks” on themselves: the serotonin they release slows down its own source. An engineer will recognize an amplifier with negative feedback here, a thermostat that holds the level. True, experiments show that this self-braking is addressed and causes only short pauses, so the thermostat role is for now more a hypothesis than a fact.

What Waiting Is Worth

What global quantity is worth keeping slow and the same for the whole network? A good candidate is *patience*. Picture a

choice: a small reward now or a big one a few seconds later. How long you are willing to wait depends on how fast the future “loses value” for you. According to one hypothesis, this is the knob serotonin turns. Some experiments support it: if serotonin neurons are switched on artificially, a mouse waits longer for a delayed reward before giving up.

This chapter introduces three devices that all the brain’s radio stations will use: the metronome neuron, a “wire” that is really a volume, and a slow level in place of separate clicks.

Formulas, models, exercises, and sources are in the full version, Chapter 5.

Chapter 6

How the Machine Learns

In all the circuits so far, the strength of the connections was chosen by an engineer. A living brain has no engineer. The connections must tune themselves, on the fly, and in such a way that the machine does something useful. That is learning.

The main constraint is very strict. Each connection changes itself, and it can know only what happens nearby: whether the sender is working, whether the receiver is working, and which substances reach it. Nobody can send a connection a personal correction. This rules out the main method for training artificial neural networks, where the error runs backward through the network and each connection gets its own correction.

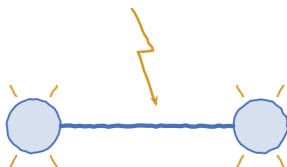
Together Means Connected

The simplest rule: if the sender and the receiver work together, the connection between them grows stronger. It is local and needs no shared timing. Such a rule, with a little forgetting added, finds the most important thing in a stream of signals: whatever varies the most. There is also a version that looks at the timing of the clicks: the connection grows stronger if the sender clicked *before* the receiver, that is, could have been the cause, and weaker if it clicked after.

But these rules have one limit: they learn what is *frequent*, not what is *useful*. A connection that sees only its neighbors does not know whether an action brought juice or pain.

The Third Signal

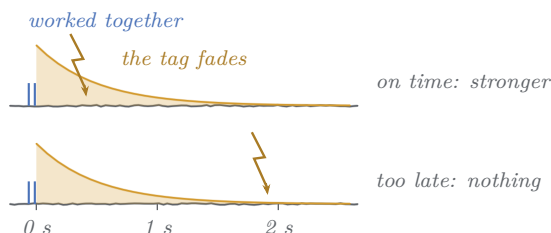
It is brought by dopamine, the “better than expected” radio station. Let a connection change only when three things coincide: the sender worked, the receiver worked, and dopamine arrived.



The connection grows stronger when both neurons worked together and a third signal arrived: “it went better than expected.”

But here is a difficulty: the reward comes not at once but a second or two later. By then the sender and the receiver have long gone quiet. The connection needs a memory that

it took part. And it has one: working together leaves a tag on the connection, like a sticker that peels off by itself in a second or two. If dopamine arrives while the sticker holds, the connection changes. If it does not, the tag disappears without a trace. This solves the addressing problem: there is one dopamine for everyone, but only the connections carrying a sticker change.



Working together leaves a tag on the connection that fades in a second or two. Dopamine changes only the tagged connections.

Noise as Search

Why does this rule lead to something better and not just anywhere? Imagine you are blindfolded and looking for the top of a hill. You take a random step, and someone shouts “higher!” or “lower!” If “higher,” you keep the step. Step by step you climb without once looking at a map. The brain does the same: neurons jitter a little at random, dopamine reports whether things got better, and the connections that took part shift in the profitable direction. The noise that got in the way of sending messages becomes a search here.

Now it is clear why dopamine reports not the reward but the *surprise*. If it simply shouted “juice!” after every

success, it would also reinforce everything the network was doing, both right and wrong. In our model such a network does run its connections up thousands of times over, and sometimes gets stuck forever on the worse choice. Subtracting the expectation makes learning fair.

The Price of One Wire

Broadcast learning has a price. One signal goes to everyone, and it mixes in the noise of all the neurons that affect the result. The bigger the network, the longer it takes each connection to pick out its own share. So dopamine apparently trains only relatively small circuits that choose actions, while the huge cortex learns mostly by local rules.

Who Computes the Surprise

One question remains: how do the dopamine neurons compute “better than expected”? A critic is needed: a group of neurons that constantly estimates how much good still lies ahead. The surprise is the reward received plus how much the expectation jumped. The lamp lights up: the expectation jumps, a burst. The promised juice arrives: there is a reward, but the expectation is used up at the same instant, no surprise. The juice does not arrive: the expectation collapses for nothing, a dip. All three cases from the first chapter come out by themselves. That dopamine behaves exactly like this is measured. How exactly the brain computes this surprise is still a hypothesis.

Formulas, models, exercises, and sources are in the full version, Chapter 6.

Chapter 7

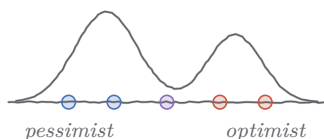
What Dopamine Really Says

In the last chapter dopamine was simple: one number, “better or worse than expected.” That is a first approximation, and it works well. But experiments in recent years have shown that more than one number runs along this “wire.” This chapter is about where the picture gets more complicated and where scientists are still arguing.

Optimists and Pessimists

Dopamine neurons are not all alike. Some respond more strongly to pleasant surprises than to unpleasant ones: these are optimists. Others do the opposite: pessimists. If each neuron learns by its own rule, the optimist learns what the reward will be at best, and the pessimist at worst. Together they record not one average expectation but the whole spread of possible outcomes, like a bookmaker who knows not only the favorite but everyone’s odds. The idea agrees with ex-

periment; why the brain needs the whole spread is still a hypothesis.



Each dopamine neuron holds its own point on the curve of possible rewards: together they remember the whole curve.

Not Only “Better” but Also “Important”

Some dopamine neurons also flare up at unpleasant but important events: a loud sound, danger. It seems the dopamine network carries not only the sign of the surprise but also its significance, a “pay attention” signal.

Two Signals on One Wire

The fast bursts teach. But the slow average level of dopamine, which changes over minutes, seems to say something else: how rich the surroundings are right now. If there is plenty of reward around, time is precious, and the animal acts faster. Experiments confirm that the dopamine level is linked to the vigor of actions. An engineer would call this separating signals by frequency: the high frequencies carry one thing, the low frequencies another.

A Ramp Near the Goal

When an animal approaches a reward, dopamine often does not flare but rises smoothly. One explanation: this is the expectation itself, a signal saying “the goal is close, push on.” Another keeps the surprise idea: from far away the animal judges its surroundings vaguely, more and more precisely as it gets closer, and each refinement is a small pleasant surprise. An elegant experiment in a virtual corridor, where the animal was instantly moved closer to the goal, favors the second: dopamine responded with a burst, as a surprise should.

Looking Back

There is also a bold alternative: dopamine answers not the question “what will happen?” but the question “what caused the reward?” It looks into the past rather than the future. Experiments in which the two theories predict different things contradict one another so far. This is a live scientific debate.

A Vector Instead of a Number

Finally, dopamine also responds to a change in the *flavor* of the reward when its value has not changed, and it helps learning without any reward at all. It seems this is not one error but a whole bundle of errors, one for each feature of what happened.

The chapter’s conclusion: most of the new theories do not overturn the simple picture but extend it. Dopamine is not one wire but a small harness of several. Only “looking back” seriously disputes it.

Formulas, models, exercises, and sources are in the full version, Chapter 7.

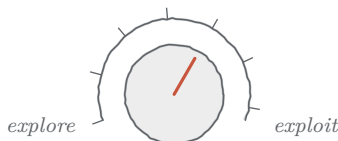
Chapter 8

The Adventure Knob

You are choosing where to have lunch. Go to the café you know, or try a new one? If you always go to the one you know, you will never find out that there is a better one around the corner. If you keep trying new ones, you will often have a bad lunch. Where is the happy medium? This problem, whether to explore or to exploit what you have found, faces every creature that learns. In the brain it is apparently handled by the **NORA** radio station: noradrenaline.

The Contrast Knob

Noradrenaline changes how sharply a neuron responds to a signal. With a weak effect the neuron is like a smooth dimmer: a little more signal, a little brighter response. With a strong effect it is like a switch: below the threshold, silence; above it, a full response. More precisely, experiments show that noradrenaline suppresses the background chatter of neurons more than their response to a real signal; the “contrast knob” is a convenient model of this effect.



The adventure knob: turn it left, and the machine tries new things; turn it right, and it firmly picks the proven one.

Weigh or Decide

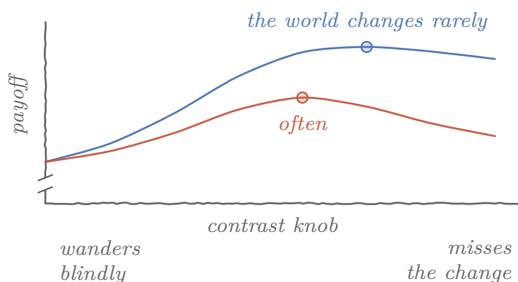
Recall the two groups from the third chapter that inhibit each other. When the contrast is low, both work, the weaker one a little more quietly: the network is still “thinking,” weighing the options. When the contrast rises above a certain threshold, the network abruptly switches to “decided”: one group wins, the other falls silent. A single knob switches the machine between deliberation and decision.

How Much to Explore

Let us add noise. At low contrast the noise easily outweighs a small difference between the options, and the machine often tries an option that is not the best: it explores. At high contrast the noise decides nothing, and the machine always takes what it considers best: it exploits.

In our model the world changes from time to time: the best option becomes the worst. The payoff depends on the knob like an upside-down letter U. Too little contrast, and the machine wanders blindly. Too much, and it fails to notice that the world has changed and stubbornly keeps going to the café that has gone downhill. And the more often the

world changes, the lower the best setting: in a changeable world it pays to explore.



The model's payoff at different knob settings. Each curve has a peak; when the world changes often, the peak is lower and further left.

Who Turns the Knob

A natural sign of change is surprise: the errors have become larger than usual. According to one hypothesis, noradrenaline reports exactly such an unexpected change. There is an important subtlety: the steady, background level of noradrenaline goes more with exploring and distraction, while short bursts at an important event go with focus and decision. Perhaps a burst works like an interrupt in a computer: “drop what you are doing, the world has changed, rebuild the plan.”

An indirect sign of this system is visible from the outside: the pupil widens when something changes unexpectedly and people start learning faster. True, the pupil tracks more than noradrenaline, so it is a hint, not an instrument.

And an honest admission: in our model, adjusting the knob by surprise gained very little over the best fixed set-

ting. Where exactly such adjustment pays off remains to be understood.

Formulas, models, exercises, and sources are in the full version, Chapter 8.

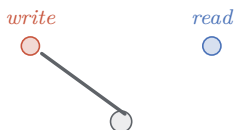
Chapter 9

Write and Read

Memory in the brain has an awkward feature: memories are stored in the same connections that do the work. There is no separate hard drive. Every time the network recalls, it uses its connections, and every time it learns, it changes them. How do you write the new without spoiling the old? A computer has a separate signal for this: “write enable.” In the brain its role seems to be played by the **ACET** radio station: acetylcholine.

Two Modes

Imagine a network that stores several pictures: show it half of one, and it fills in the rest. That is reading. Now show it a new picture that resembles an old one. If the network fills it in out of habit, it will see the old rather than the new, and learn an error. To write the new, the network must stop trusting its memory for a while and look at the world.



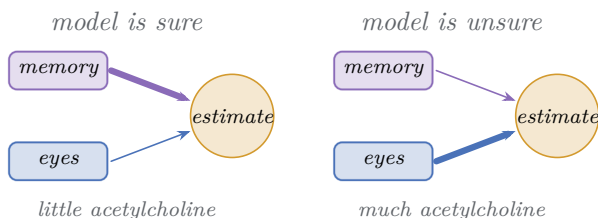
One wire switches the network: write the new, or recall the old.

Acetylcholine does exactly this, and all three of its effects are measured: it weakens the network's internal connections, through which it "recalls," strengthens the signals coming from the sense organs, and makes the connections easier to change. Plenty of acetylcholine means write mode: the network listens to the world and learns. Little means read mode: the network relies on its memory and fills in.

In our model there is no level at which both go well. For writing, the internal connections must be muted; for reading, they are needed at full strength. So a switch is necessary.

How Much to Trust Your Eyes

Engineers solved a similar problem long ago. When a ship holds its course, the navigator has a calculation of where the ship should be, and has new but imprecise observations. How much to trust each? The optimal filter answers: trust the observations the more, the less sure you are of the calculation. And the same recipe contains a second point: the less sure you are, the faster you must relearn. One knob controls both the weight of the input and the learning rate. That is exactly what acetylcholine does. Hence the hypothesis: it tells the network how uncertain its own model of the world is.



How much to trust your eyes. If the network is sure of its model of the world, the estimate comes almost entirely from memory. If it is unsure, from the observations, and then the network also relearns faster. According to the hypothesis, acetylcholine turns this knob.

Division of Labor

Recall the last chapter. When the world suddenly changes, noradrenaline, according to the hypothesis, notices the surprise. And acetylcholine keeps the network in write mode until the new situation is learned. One reports “something has changed,” the other “write.”

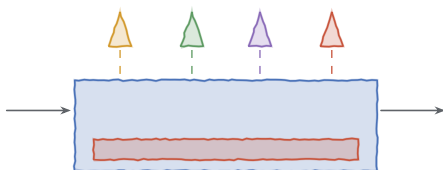
Rewriting at Night

During deep sleep the acetylcholine level is low. The network is cut off from the world and is in read mode. It is exactly then that the day’s events are “replayed” in it; this is measured. That what was learned during the day is being rewritten into long-term memory at this time is a plausible hypothesis, which we will return to in the chapter on sleep.

Formulas, models, exercises, and sources are in the full version, Chapter 9.

Chapter 10

The Whole Machine



The whole machine: the data bus, the brakes in it, and four radio control channels above.

We took one type of neuron at a time and asked what it adds to the machine. It is time to put the parts together.

The result is a machine of three layers. At the bottom is the huge telephone network of **GLUT** neurons: the data run through it. Next to it are the braking **GABA** neurons: they control the flow and decide what gets through, when, and how loudly. And above it all are a few radio stations, each with its own knob. **DOPA** says which changes in the connections to keep. **SERO** holds the overall level and, according to the hypothesis, the horizon of patience. **NORA** chooses between exploring and deciding. **ACET**, between writing and reading. Each radio station is really a small bundle of channels, not

one wire, but there is still only a handful of channels for eighty-six billion neurons.

One Story

Let us follow one event through the whole machine. An animal has long known: press lever A and you get juice. One day the world changes: A no longer works, and the juice now comes from lever B, which the animal has hardly used.

No juice comes after A: dopamine answers with a dip, and the connections that chose A weaken. The dips repeat, the errors become larger than usual, and noradrenaline, according to the hypothesis, reports: the world has changed. The contrast knob goes down, the choice becomes softer, and the noise more often nudges the animal to try B. Acetylcholine switches the network into write mode. Trying B brings unexpected juice: dopamine flares, and the connections that chose B grow stronger. After a few attempts the expectation catches up with the new world, the surprises end, and the knobs return to their places.

Note that no knob knows what the others are doing. Each responds to its own cues, and the sequence of actions assembles itself, as in an asynchronous circuit where each block fires when its inputs are ready. That each knob behaves this way on its own is mostly measured. That they work together in concert is still a hypothesis.

How It Differs from Your Laptop

A laptop has a common clock; the brain does not. The elements of a laptop are reliable; a synapse loses more than half of its signals. A laptop's memory is separate from its

processor; in the brain it sits in the same connections. A laptop learns (if it learns) by backpropagation of error; the brain, by local rules and a single broadcast “better than expected.” And most important: a laptop spends tens of watts on one processor, while the brain spends twenty watts on everything.

What We Don’t Know

We do not know how much the cortex uses the precise timing of clicks. We do not know how it tunes billions of connections in many-storied circuits when it has one teacher for all. We do not fully understand what the radio stations transmit. We do not know how distant parts of the brain coordinate their work with no common clock. And so far nobody knows how to build a machine that would do the same on twenty watts.

From here the book goes beyond this core: to the inputs and outputs of the machine, to debugging it, to sleep, and to machines made of living neurons on a chip.

Formulas, models, exercises, and sources are in the full version, Chapter 10.

Chapter 11

How the World Gets In

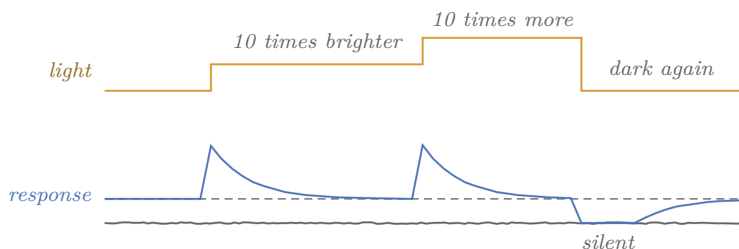
Every sense organ is a transducer, like a microphone or a photocell. Light, sound, or a molecule acts on a special protein in a sensor cell, the protein opens a “tap,” current flows, and at the end of the chain come clicks that go out into the network. Almost all sensors release glutamate, so the machine’s inputs plug straight into its telephone network.

At the Limit of Physics

The brain’s sensors work at the very limit of what is possible. A light sensor in the dark responds to a single particle of light. A sound sensor notices a displacement of less than a nanometer, smaller than a few atoms. In dim light the precision is limited no longer by the eye but by light itself: the particles arrive at random, and to tell a shade apart you have to collect more of them. That is why we see more slowly at dusk: the eye collects light longer, like a camera with a long exposure.

Automatic Exposure

Illumination varies a billionfold from a starry night to sunlit snow, and loudness even more. Yet the rate of clicks varies from zero to a few hundred per second. Squeezing one into the other directly is impossible. The solution is the same as in a camera: automatic exposure. The sensor constantly adjusts its sensitivity to the average level around it and reports not the brightness but the brightness *compared with the background*. Anything constant stops evoking a response over time, while every change evokes one. From the very start, the machine's inputs talk about changes.



The light gets brighter in steps, each tenfold. The sensor answers each change with a burst and quickly returns to its former level. When it gets dark again, it falls silent for a while.

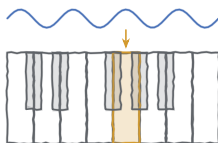
Engineers have copied this trick in event cameras. Such a camera does not take frames on a clock: each of its pixels reports “brighter” or “darker” on its own when something changes. It is the same pair of “yes” and “no” wires from the second chapter, only in silicon.

The Eye Compresses the Picture

The eye has about a hundred million light sensors, and about a million wires carry the image to the brain. Before the picture leaves the eye, it is compressed about a hundredfold. Flat areas cost almost nothing; edges and changes are sent. By an estimate made on animal eyes and scaled to humans, the eye sends the brain about ten million bits per second, like an old network cable.

The Ear Is a Piano

The ear splits sound into frequencies, the way an engineer would set up a bank of filters. An elastic partition in the ear responds to different frequencies at different places, and the sensors on each patch listen to their own “key.” The number of the sensor that fires is the pitch.



The ear is built like a piano in reverse: each key listens for its own note.

What is more, the ear is not passive. Some of its cells themselves rock the partition in time with the sound and amplify quiet sounds thousands of times, while hardly amplifying loud ones. This is an amplifier right at the edge of self-oscillation: that is where it is most sensitive. And at low frequencies the neurons also click in time with the wave, preserving the exact timing of the sound; that is how the brain finds the direction of the source. The higher the

frequency, the worse the neurons keep up with it, and only the place code remains.

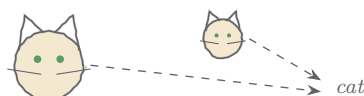
The Other Senses

In almost every sense you can recognize a trick from the earlier chapters. Temperature is carried by two wires, separate ones for warmth and for cold. Touch uses sensors that respond to contact and fall silent at once, and sensors that keep responding as long as the pressure lasts. And smell works like a barcode: humans have about four hundred types of smell sensors, and an odor is the set of types that fired. The number of such sets is astronomical.

Formulas, models, exercises, and sources are in the full version, Chapter 11.

Chapter 12

How to Recognize a Cat



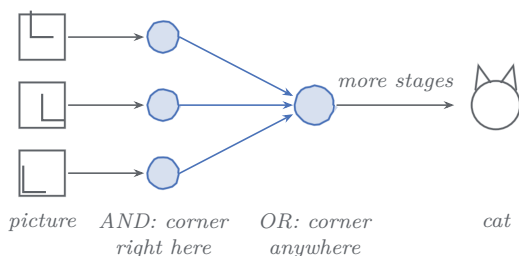
A big cat and a small one are completely different pictures on the eye's sensors, yet the answer is the same.

A cat in a picture can be on the left or on the right, near or far, in the sun or in the shade. Each time a completely different pattern falls on the eye's sensors, yet you recognize it in a fraction of a second. That is the difficulty of recognition: the same thing has to be recognized in thousands of different views.

The Assembly Line

The brain solves the problem with an assembly line of about ten stages, and the signal passes through it in about a hundred and fifty milliseconds. Each stage does two things. First,

“and”: a neuron fires if the right details stand in the right arrangement, say, two segments meet in a corner. Then, “or over all places”: the next neuron fires if there is a corner anywhere nearby, no matter exactly where. The first operation makes recognition discerning, the second makes it tolerant of shifts. Stage by stage, details add up to parts and parts to objects, while shifts and scales are forgiven.



One stage of the assembly line: “and” assembles a corner in one place, “or” forgives exactly where it stands.

This is exactly the scheme copied by the convolutional neural networks that recognize pictures in your phone. And they return the favor: networks trained to recognize objects predict fairly well how neurons at the upper stages of vision respond.

Learning from Video

But who taught the brain? Nobody showed you a million pictures labeled “cat.” A plausible answer is video. The world flows continuously, and within a fraction of a second an object hardly changes, while its position and lighting do. If connections strengthen between what is seen now and what was seen a moment ago, the network will link different views of one object by itself: what is slow is the object, what is

fast is the circumstances. There is an experiment in which objects were quietly swapped during eye movements, and the neurons began to treat different objects as one. As a general rule this is still a hypothesis.

Illusions Are Fingerprints of Mechanisms

Illusions are not breakdowns but traces of how the machine is built. If the eye looks at a waterfall for a long time and then at motionless rocks, the rocks “crawl” upward: the pair of “down” and “up” channels is thrown off balance because one of them is tired. The famous dress that some see as white and gold and others as blue and black divides people by the lighting their brain silently assumes. The drawn “rotating snakes” seem to move because bright patches are processed slightly faster than dim ones, and the time difference is read as motion. And a cube that pops out and then sinks in is two groups of neurons competing with each other; according to one model, the switch in outcome is decided mostly by noise.

Curiously, a neural network trained to predict the next frame of a video also “sees” the snakes rotate. So some illusions are the unavoidable price of being able to predict.

Brain versus Neural Network

The resemblance is not complete. The brain needs few examples, learns almost without labels, and spends twenty watts on everything. A neural network can be fooled by a doctored picture that a human would not notice. True, such doctored pictures also throw people off a little if they are flashed for an instant. Where the real boundary between these two

machines lies is still being worked out.

Formulas, models, exercises, and sources are in the full version, Chapter 12.

Chapter 13

The Only Fast Output

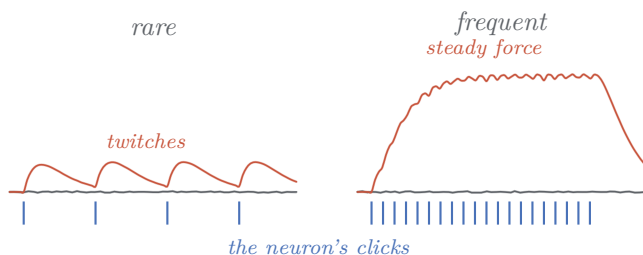
Everything you do quickly in the world—a word, a glance, a step—is a contraction of muscles. The machine also has a second, slow output, a chemical one; Chapter Sixteen is about it. Here we deal with muscles.

The last link in the chain is formed by the **ACET** neurons. Each muscle fiber listens to exactly one such neuron, and one neuron controls a group of fibers, from hundreds to a couple of thousand. Interestingly, the junction between nerve and muscle is the most reliable connection in the body. Inside the brain, connections lose more than half of their signals, but here each click releases several times more substance than needed. At the output an error costs a movement, and reliability is bought with a margin.

A Drumroll Becomes a Push

One click, one short twitch of the muscle. If the clicks come rarely, the muscle jerks in separate twitches. If they come often, the twitches merge into a steady force, the way a

drumroll merges into a rumble. The muscle turns a stream of spikes into force, like the simplest digital-to-analog converter.



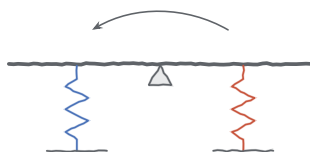
Rare clicks give separate twitches of the muscle; frequent ones merge into a steady and much larger force.

Floating-Point Force

The brain has two force knobs: how often the neurons click and how many groups of fibers are switched on. The groups switch on in order, from the weakest to the strongest, and nobody computes this order: small neurons are simply easier to excite. As a result, each new group adds roughly the same *fraction* to the force already there. When you pick up an egg, the force is metered out in tiny portions; when you lift a suitcase, in large ones. An engineer will recognize a floating-point number here: the exponent is set by the number of groups switched on, and the exact value by the rate of clicks. The flip side: the stronger the movement, the less precise it is. According to one influential hypothesis, this is why our movements are so smooth: a smooth command gives the least scatter.

Pull, Don't Push

A muscle can only pull. So each joint is served by a pair of muscles pulling in opposite directions: again two wires instead of a sign. The difference of their forces turns the joint, and their sum makes it stiffer. When you carry a full cup, you tense both muscles at once: the rotation is the same, but the arm takes a bump more firmly.



A muscle can only pull, so there are two: the difference of the forces turns the joint, the sum makes it stiffer.

Feedback with a Delay

Muscles have stretch sensors, and the shortest loop closes right in the spinal cord: stretch a muscle, and it contracts by itself, while the opposing muscle is switched off for that time. It is switched off by a braking neuron of the **GLYC** type: so this type has found its job.

But every loop has a delay: tens of milliseconds in the spinal cord, hundreds through the eye. And delayed feedback makes a system swing, like a driver who steers while seeing the road with a delay. The longer the loop, the more slowly it must correct. The limit beyond which the spinal loop starts to swing is about eight oscillations per second. This is close to the frequency of the fine tremor that every healthy person has, and the stretch loop is one of its likely sources.

So how do we move fast and precisely? According to a widespread hypothesis, we predict: we keep an internal model of the body and compute the result of a command without waiting for the sensors.

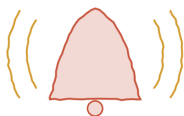
Rhythm Without a Metronome

Walking and breathing are rhythmic, but they need no metronome. Two groups of neurons that inhibit each other and gradually tire take turns by themselves: the winner tires, the rested rival takes over, and so on again and again. A constant command from above, “walk,” turns on the spot into a “left, right” rhythm.

Formulas, models, exercises, and sources are in the full version, Chapter 13.

Chapter 14

The Alarm Signal



Pain is a fire alarm, not a thermometer.

All the other senses report what the world is like. Pain does not. It reports that something threatens the body, and its loudness is set largely by the machine itself, not by the stimulus. Pain is a fire alarm, not a thermometer.

Pain sensors fire only at something strong: at heat above forty-odd degrees Celsius, at heavy pressure, at caustic substances. They adapt much less than touch sensors do: an alarm must not fall silent while the threat is still there.

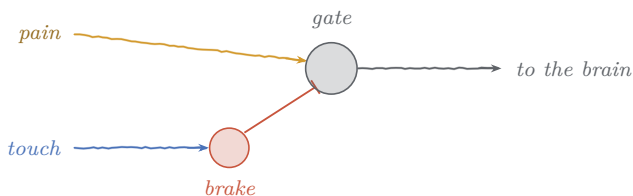
Two Wires

Remember hitting your finger. First comes a sharp, precise pain: where and when. A moment later, a dull, spread-out,

lasting one: how bad. These are two different wires. One is fast; its signal arrives in tens of milliseconds. The other is slow; its signal takes on the order of a second.

The Gate

Why do we rub a bruise after bumping it? In the spinal cord the pain signal passes through a “gate.” Touch wakes a braking neuron that half-closes it. So rubbing a bruise really does help, but only against moderate pain; strong pain gets through anyway. That strong pain itself switches off the braking neuron is part of the original gate theory, and there is no direct confirmation of it yet.



The gate in the spinal cord. The pain signal goes to the brain through a gate neuron, and touch wakes a braking neuron that half-closes the gate. So rubbing a bruise helps, but only against moderate pain.

A Volume Knob from Above

The brain can make pain louder or quieter by itself. Radio channels run down to the gate from the brain above: **SERO**, **NORA**, and special substances resembling painkillers that the brain makes itself. In a dangerous situation pain can almost vanish: someone running from danger has no time

to limp. Expecting relief also dulls pain, partly through the same substances.

When the Alarm Gets Stuck

Repeated pain raises sensitivity: the same blow starts to hurt more. This is learning too: a loop that turns itself up while the wound heals. But a strong loop has a danger: it can “stick” and keep ringing after the wound has healed, like the group of neurons from the second chapter that keeps itself going.

Teacher and Interrupt

Pain is also a teacher: for dopamine’s “better or worse than expected,” it works as a negative reward. And it interrupts what you are doing, like a burst of noradrenaline: drop everything, deal with the danger. And the fastest reaction, pulling your hand away from something hot, closes right in the spinal cord through that same pair of opposing muscles from the last chapter, before you have even had time to feel the pain.

Formulas, models, exercises, and sources are in the full version, Chapter 14.

Chapter 15

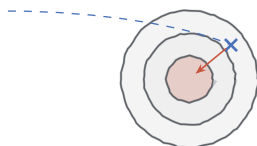
How the Machine Learns to Move

We have seen two ways of learning. Without a teacher, “together means connected,” the machine remembers what occurs often. With dopamine it learns from surprises: things went better or worse than expected. But when you miss while throwing a ball at a basket, you know more than that things went badly. You know *by how much* and *in which direction* you missed. This is the third teacher: an error with a direction, personal to each output.

The Error Wire

The brain has a separate circuit for learning movements, and it is built in a very unusual way. A huge number of small **GLUT** neurons, tens of billions, spread the input signals out into a very long list of features. The output neurons of this circuit are braking **GABA** neurons; there are tens of millions of them, and each listens to a hundred thousand inputs. And

to each output neuron runs exactly one special wire: the error wire. If an error arrives while an input was active, that input weakens. This is the classic “reduce the error” rule that engineers use to train adaptive filters.



A miss reports not only “bad” but also “by how much and in which direction.”

Such a teacher is faster than the dopamine one: each output has its own error, and there is no need to fish its share out of one signal common to all. True, the error arrives late, and the connection again needs a “sticker” that remembers it took part. Experiments show that the lifetime of this sticker is matched to the delay of the error.

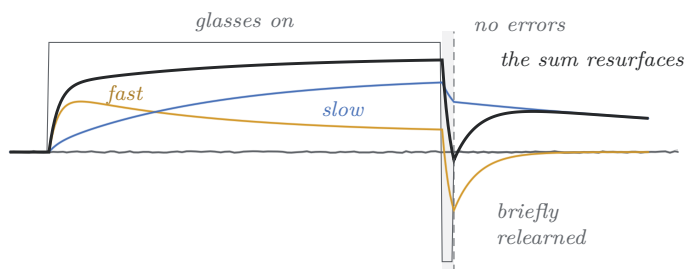
The Internal Model

What does such a circuit learn? A model of its own body: what result a command will produce. With it there is no need to wait for the lagging sensors; the result can be predicted in advance. According to one hypothesis, this is why you cannot tickle yourself: the brain knows in advance what will happen and subtracts the prediction.

A Fast and a Slow Learner

Put on glasses that shift the picture sideways, and throw darts. The first throws go off to the side, but within dozens

of tries you adapt. Take the glasses off, and you miss again, now to the other side. The interesting part comes next. The experiments are described well by two learners at once: the fast one learns fast and forgets fast, the slow one learns slowly but remembers long. If after a long adaptation you briefly relearn the other way, the total correction returns to zero, but the old skill then partly resurfaces by itself: the fast learner has forgotten, but the slow one remembers. A model with one learner cannot do this, but people can.



The correction of the two learners in the model. After brief relearning the sum is at zero, but the slow learner remembers the old skill, and it resurfaces by itself.

Why two learners? According to the hypothesis, it is the same recipe as the navigator's from Chapter Nine: the world changes both fast and slowly, and it is wise to track both.

Formulas, models, exercises, and sources are in the full version, Chapter 15.

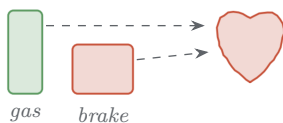
Chapter 16

The Second Output

The machine has two outputs. The fast one is the muscles. The slow one is the body's chemistry: the brain controls the heart, the blood vessels, and the glands, and sends messages through the blood to the whole body at once. The blood is the most broadcast bus of all: one message reaches every cell in about a minute, and, as always, its meaning is given by the receiver's "lock."

Gas and Brake

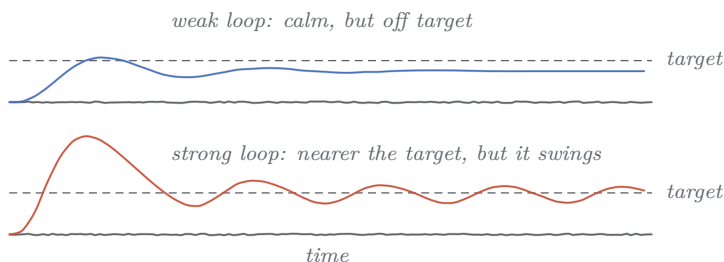
The heart beats by itself; it has its own metronome. The brain only adjusts the tempo with two wires of opposite sign. **NORA** is the gas pedal: it speeds the heart up, but slowly. **ACET** is the brake: it slows it down fast, because it does not need a long chain of chemical reactions. And here at last the **ADRE** type finds a job: adrenaline released into the blood is the same gas pedal, only pressed for the whole body at once.



The heart is driven by two wires of opposite sign: gas and brake.

A Cascade with Feedback

Many hormones are controlled by a cascade: the brain orders the first gland, that gland orders the second, the second secretes a hormone, and the hormone tells the brain there is already enough of it. This is a three-stage thermostat. An engineer knows what is dangerous about such a circuit: if the loop is made too “sensitive,” it starts to swing, and the level no longer holds but oscillates. So the precision of such a regulator is limited: it cannot be made both stable and very precise at the same time.



A model of a three-stage hormone cascade. Weak feedback settles the level but leaves it well below the target. Strong feedback brings it closer, but then the level no longer holds; it swings.

Pulses, Not a Level

Some hormones work only in pulses. If they are supplied constantly, the receiving gland “tires” and shuts down; if in short pulses once an hour, it works. The receiver reads not the level but the rhythm, like the synapse from the Morse code chapter that hears only changes.

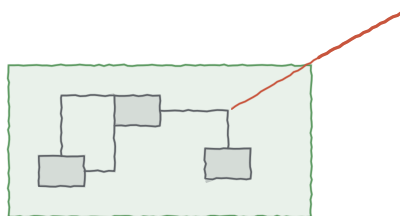
The Day

Finally, the body has a slow oscillator with a period of about a day. Its natural period differs slightly from twenty-four hours, and every morning light tunes it, the way a radio receiver tunes to a station. Do not confuse it with a computer’s clock generator: it does not measure out the steps of a computation. It switches the operating modes of the whole machine: day and night, waking and sleep.

Formulas, models, exercises, and sources are in the full version, Chapter 16.

Chapter 17

How to Fix a Machine Without a Schematic



A board without a schematic: the probe hears only a few wires out of billions.

Imagine you are handed a working circuit board without a schematic and asked to repair it. An engineer has four tools: a probe to listen to a signal; a signal generator to feed in a signal of one's own; the option of disconnecting a part and seeing what breaks; and access to the control registers. The brain is studied with exactly these tools.

A Probe for a Thousand Neurons

An electrode stuck into the brain hears the clicks of a few neurons nearby. It would seem that a thousand electrodes are nothing against eighty-six billion. Yet they are enough to control a cursor on a screen. Why? Neighboring neurons have similar noise, and the activity that controls a hand is really described by just ten or twenty shared variables. There is no need to listen to everyone; it is enough to hear these few “voices.”

The raw signal from a thousand electrodes is hundreds of millions of bits per second: too much for a radio link from under the skull. The clicks themselves carry about a thousand times less. So it pays to pick out the clicks right on the implant and send only them outside: the same compression trick as in the eye.

What Neuralink Does

The company Neuralink took on the engineering side of the problem: thin flexible threads instead of rigid needles, a robot that implants them, and a sealed wireless implant. Its published paper describes a system connected to the computer by a cable, and names compression and wireless communication as plans. The data on a person with paralyzed hands who controls a cursor are so far known mainly from the company’s own reports; according to them, the implant has about a thousand channels on a few dozen threads, and some threads moved after implantation. The idea itself is not new: academic groups with rigid arrays of about a hundred electrodes have already let people control a cursor, “write” letters by thought, and turn attempts to speak into text on a screen at about sixty words per minute.

Reading and Writing

The decoder reads the click rates of many neurons and gets less than ten bits per second, a tiny fraction of what the neurons themselves carry. On the other hand, the brain and the decoder learn from each other: the person adapts to the decoder, and the decoder to the person. Two learners adapting to each other are a delicate problem: even when each is stable on its own, together they can start to swing.

Writing into the brain is harder than reading. Current through an electrode excites everything around it, not the one neuron you want. You can “draw” dots of light, like glowing pixels, and a person will make out simple letters. But nobody yet knows how to write the eye’s compressed code so that the brain sees a picture.

What the Probe Does Not See

An electrode hears the telephone network of data. But the radio stations, dopamine, serotonin, noradrenaline, and acetylcholine, are concentrations of substances, and the probe does not hear them. Nor does it see the strengths of the connections, where all the memory is stored. And it does not see pain as a person feels it. A debugger that sees the data but not the registers will always wonder why the same input gives different answers.

Formulas, models, exercises, and sources are in the full version, Chapter 17.

Chapter 18

A Neuron Is Not One Device

In the first chapter we sorted neurons by the substance they release. An engineer would sort them differently: by what device each one works as. And it turns out that behind the same appearance hides a whole box of electronic components.

A Box of Components

The leaky bucket from the second chapter is a leaky integrator: it adds up signals over a short window. With a very small window it becomes a coincidence detector. A neuron that responds only to changes and falls silent at once is a filter that passes changes. A neuron that runs by itself is a metronome, an oscillator; if an input adjusts its tempo, it is a voltage-controlled oscillator. A neuron that puts out bursts of clicks is a burst generator, writing “dashes.” A neuron that clicks in time with a sound wave is a phase detector.



Behind the same appearance hides a whole box of components: a resistor, a capacitor, a coil, an oscillator.

There are also devices we have not talked about before. A *resonator* is like a swing: it responds best to pushes of a certain rhythm, like a swing that goes higher only if you push it at the right moment. Inside the neuron the role of the spring is played by a slow current that resists change. An *integrator without leak* holds a value for a long time, like a capacitor that does not leak; this is how the network that holds the position of the eyes works when you look to the side. But for this the feedback must be tuned almost perfectly, and that is a costly adjustment. A *latch* is a neuron that a short push switches into lasting activity until it is switched off, like a latching switch; in the neurons that control muscles, serotonin turns this mode on.

A Chameleon Device

The main discovery of this sorting: these are not different parts but different *modes* of one and the same part. The same neuron can be an integrator by day and a burst generator by night, depending on what the radio stations tell it. An engineer would call the brain a field-programmable gate array: one circuit, while the settings change what the parts do.

That neurons can work in all these modes is measured. How the brain uses mode switching for computation is mostly a hypothesis.

Formulas, models, exercises, and sources are in the full version, Chapter 18.

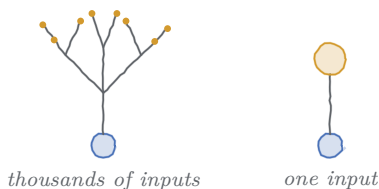
Chapter 19

How Many Inputs Are Needed

A neuron has branching input processes: dendrites. In some neurons they form a whole tree with thousands of contacts; in others there are one or two branches or none at all. For an engineer the number of inputs is a circuit parameter, and it is chosen to suit the task.

From Zero to a Hundred Thousand

Touch and pain sensors have no dendrites at all: the wire simply runs from the skin to the spinal cord, and the sensor sits at its very end. This is a line with a sensor at the far end.



Many inputs make an adder; one huge input makes a fast and precise relay.

The neurons of smell, and the neurons that pass on the signal from the sensors of the eye and the ear, have one dendrite. One input, one output: a repeater, a buffer.

The sound-direction detectors have two dendrites pointing in opposite directions, one for each ear. By putting the two inputs on different branches, the neuron makes its “and” stricter: two portions of current in one branch would get in each other’s way.

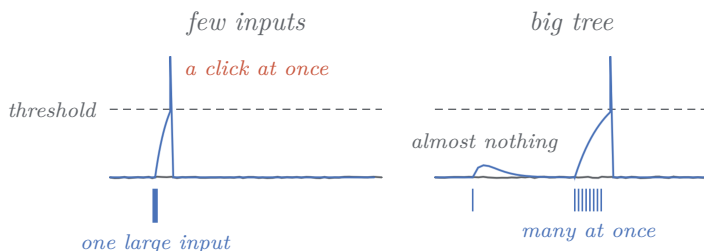
The small neurons of the circuit for learning movements have about four short dendrites, each grabbing its own input. There are tens of billions of such neurons, and each adds up only four signals. But together they spread the input out into millions of combinations, and a large output neuron with a hundred thousand inputs picks the right ones among them.

And finally, the neurons with huge trees and thousands of inputs are adders and classifiers, whose individual branches work almost like independent small circuits.

Why So

Charging decides everything. A small neuron with few inputs is charged by one large connection in a fraction of a millisecond: such a neuron is precise in time and reliably

passes the signal on. A big tree will never be charged by one connection: the charge leaks away first. It needs dozens of signals arriving together, but in return it can add them up and compare them. Few inputs and large connections are for precise timing. Many inputs are for counting.



A small neuron is charged by one large connection in a fraction of a millisecond and clicks at once. A big tree hardly changes from one input: it needs dozens of signals arriving together.

This gives us a third sorting of neurons: by substance, by device, and by number of inputs. Each sees the same machine from a new side.

Formulas, models, exercises, and sources are in the full version, Chapter 19.

Chapter 20

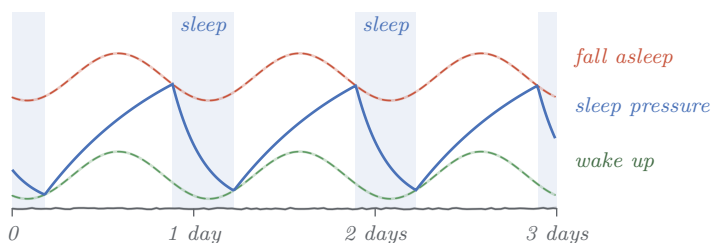
What the Machine Does at Night

Sleep is not the machine switching off but a change of mode. You can see it in the radio stations. In deep sleep almost all of them are turned down: acetylcholine, noradrenaline, serotonin. In dreaming sleep acetylcholine returns to its daytime level, noradrenaline and serotonin are almost silent, and the output to the muscles is switched off by the brakes; that is why we do not actually run when we run in a dream. All of this is measured.

When It Is Time to Sleep

Imagine a capacitor that charges all day while you are awake and discharges at night. The charge is the “sleep pressure.” You fall asleep when the charge reaches an upper threshold and wake up when it drops to a lower one. And the thresholds themselves drift smoothly up and down with the daily rhythm we met earlier. Such a simple circuit arrives by itself at

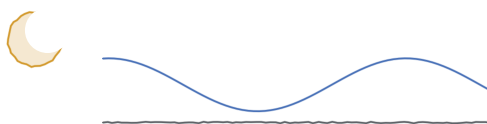
sixteen hours of waking and eight hours of sleep. It also explains why after a sleepless night we sleep not twice as long but more deeply: a full capacitor discharges faster. A candidate for the “charge” is a substance that builds up in the brain as it works; that it is exactly this substance is still a hypothesis.



Sleep pressure builds up by day and melts away at night; we fall asleep at the upper threshold and wake up at the lower one, and both thresholds swing with the time of day.

A Slow Swing

In deep sleep whole regions of the brain switch on and off in sync, less than once a second. This is a familiar circuit by now: a group that excites itself and tires is an oscillator. By day acetylcholine suppresses the fatigue of neurons, and the same group works steadily; at night there is little of it, and the network starts to swing.



At night the network swings slowly: at one moment almost all the neurons work, at the next almost all are silent.

Fast Forward

The most interesting thing happens within these swings. Sequences of neuron firings that happened during the day are “replayed” anew, and several times faster: about twenty times faster in the memory region, a few times faster in the cortex. If the alignment of these replays with the slow swings is strengthened artificially, memory improves; this is a causal experiment. What is it for? A plausible hypothesis: the slow long-term memory learns from replays mixed in with old material, so that the new does not overwrite the old. Engineers know this trouble: a neural network trained on a new task forgets the old one unless the old examples are repeated.

Pruning the Connections

Another measured fact: after sleep the contacts between neurons become smaller on average, by a few to a few dozen percent, in proportion to their size. As if all the volumes were turned down a little with one motion: the ratios are kept, what was learned stays, and room is freed for tomorrow’s learning. That this is the main task of sleep is a hypothesis.

Dreams and Cleaning

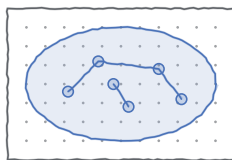
Dreaming sleep resembles a bench test: the machine runs at full power, but the inputs are replaced by internal signals and the output is disconnected. What this is for, we do not know. The idea that during sleep the brain is “flushed” of waste is still disputed: there are experiments both for and against.

Most surprising of all: sleep can be local. After long waking, separate patches of the cortex of a waking person “fall asleep” for an instant, going silent in the middle of their work.

Formulas, models, exercises, and sources are in the full version, Chapter 20.

Chapter 21

A Computer Made of Living Cells

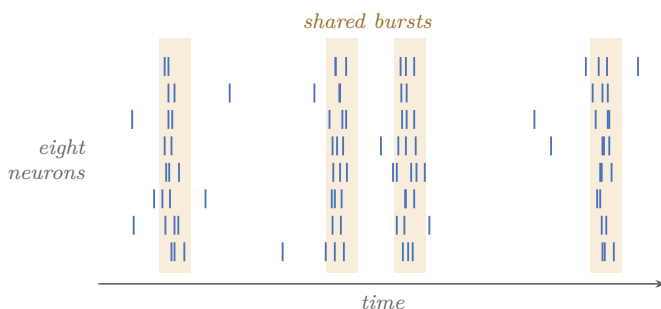


Living neurons on a chip: under them are thousands of electrodes, and each can both listen and speak.

Can a machine be built not from silicon but from living neurons? It can, and such machines are already being built. Neurons are grown right on a chip with thousands of electrodes, as a flat layer or as little clumps that resemble a tiny brain. Each electrode can both listen and deliver a signal. This is a test bench with an open circuit: everything can be measured, and everything can be tampered with. Only there are no radio stations in it: it is a data bus without control registers.

What the Network Does on Its Own

Left to itself, a culture flares up as a whole from time to time: almost all the neurons click together and then fall quiet, at intervals from a second to minutes. This is the familiar oscillator: the network excites itself and tires until the connections restore their supply of substance.



A culture on a chip left to itself. Single clicks are rare, and from time to time almost the whole network flares up at once, then falls quiet until the connections restore their supply.

A Teacher Without Dopamine

How do you teach such a network if there is no dopamine? The first methods were electrical. For example: stimulate the network until it gives the desired response, and stop at once. The response gets fixed. In another experiment a culture controlled a simple virtual “body” and learned to steer it toward a goal. In all these experiments the teacher is a pattern of current chosen by an engineer.

Dopamine by a Flash of Light

Once, researchers tried to make the teacher chemical. On one of the platforms with living clumps of neurons, dopamine was added to the nutrient fluid in a light-sensitive “cage” that keeps it from acting. When the network responded to a stimulus more strongly than before, it was “rewarded”: a flash of ultraviolet light broke the cage, and the dopamine got out. Over two hundred forty repetitions the responses grew overall. But the authors themselves honestly write that one such experiment cannot tell whether dopamine was the cause.

A Pole on a Cart

In a recent experiment clumps of neurons were taught to balance a pole on a cart, a classic problem for learning programs. Which training pulses to deliver was chosen by a program. The signals chosen by the program worked better than random ones, but after three quarters of an hour of rest what had been learned disappeared. And if the **GLUT** connections were blocked, there was no learning at all: what is learned lives right there. The teacher, however, is still outside; only now it is not an engineer but a program.

A Living Reservoir

There is another approach: the living network is not taught at all. It is used as a “reservoir,” a rich, tangled system whose responses a simple electronic readout learns to sort out. In this way a living clump helped tell voices apart. True, according to the authors, the connections in the clump itself

also changed during the work.

The Cost and the Question

A million neurons on a chip consume about a ten-thousandth of a watt, far less than the electronics around them. And the bench cannot decide by itself what counts as success: someone outside always plays that role. And as such systems grow, an ethical question arises that nobody has settled yet.

Formulas, models, exercises, and sources are in the full version, Chapter 21.

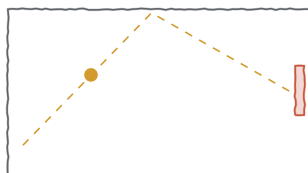
Chapter 22

Neurons Play Tennis

The most famous experiment with living neurons on a chip is DishBrain, the “brain in a dish.” About a million neurons grown on a chip with twenty-six thousand electrodes played the simplest computer tennis: a ball flies across the screen, and a paddle moves up and down.

How the Game Works

The ball enters the network through eight electrodes. Which of them clicks tells the height of the ball. How often it clicks tells how far away the ball is: four times a second when it is at the far wall, and more and more often, up to forty, as it approaches the paddle. The output is two regions of electrodes: clicks in one move the paddle up, in the other down. Every ten milliseconds the computer counts the clicks and moves the paddle. These ten milliseconds are the clock of the bench’s computer, not of the culture: the network itself works asynchronously, like any network of neurons.



The ball, the paddle, and a million neurons that move it.

The Teacher

No dopamine at all. After a successful hit the network gets a short and completely predictable signal: all eight electrodes at once for a tenth of a second. After a miss, four seconds of disorderly stimulation at random places and random moments, then a pause, and a new serve.

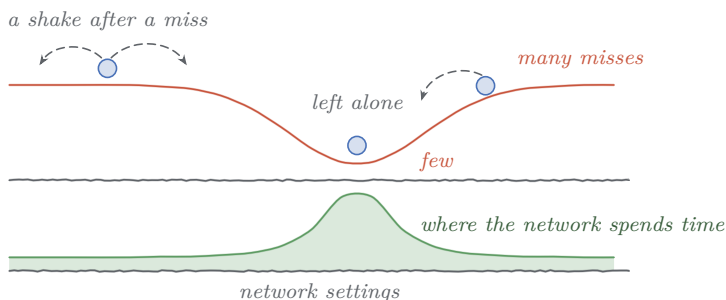
What Came Out

Over a twenty-minute session the rallies of the living cultures grew longer, and the missed serves fewer. This worked best with full feedback. When plain silence followed hits and misses instead of the signals, there was improvement too, but weaker; without feedback there was none. Human neurons played a little better than mouse neurons. But learning stretched over several days did not prove reliable.

Why the Network Learns

The authors explain the result this way: the network strives to make its sensations predictable. A miss brings chaos, a hit brings order, and the network changes so that there is more

order. But there is also a more down-to-earth explanation that also fits the data: “shake after a miss, leave alone after a hit.” Each shake scrambles the connections a little at random. Settings at which the network often misses keep getting scrambled, while the successful ones are left untouched. The network lingers by itself where there are few misses: nobody taught it; it simply stops being shaken. In our simple model such a rule does raise the share of hits.



A miss shakes the network's settings at random; a hit leaves them alone. Where misses are many, the network does not linger. Where they are few, the shaking stops, and it stays there, though nobody led it there.

How can the two explanations be told apart? An experiment is needed that has not been done: after a miss, deliver not a chaotic but a *predictable* signal of the same duration and strength. If the network learns even so, the point is not predictability but simply the difference between “after a hit” and “after a miss.” This experiment is still waiting for someone to run it.

This is the end of our journey. The twenty-watt machine is assembled from simple parts, but how exactly it puts them together into thought we so far understand only in part. Perhaps you will write the next chapter.

*Formulas, models, exercises, and sources are in the full version,
Chapter 22.*